Taylor & Francis
Taylor & Francis Group

Econometric Reviews

ISSN: 0747-4938 (Print) 1532-4168 (Online) Journal homepage: www.tandfonline.com/journals/lecr20

In-Sample or Out-of-Sample Tests of Predictability: Which One Should We Use?

Atsushi Inoue & Lutz Kilian

To cite this article: Atsushi Inoue & Lutz Kilian (2005) In-Sample or Out-of-Sample Tests of Predictability: Which One Should We Use?, Econometric Reviews, 23:4, 371-402, DOI: 10.1081/ETC-200040785

To link to this article: https://doi.org/10.1081/ETC-200040785

Published online: 06 Feb 2007.

Submit your article to this journal

Article views: 3943

View related articles

Citing articles: 59 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=lecr20

ECONOMETRIC REVIEWS
Vol. 23, No. 4, pp. 371–402, 2004

In-Sample or Out-of-Sample Tests of Predictability: Which One Should We Use?

Atsushi Inoue[1,*] and Lutz Kilian[2,3]

[1]Department of Agricultural and Resource Economics, North Carolina State University, Raleigh, North Carolina, USA
[2]Department of Economics, University of Michigan, Ann Arbor, Michigan, USA
[3]CEPR, UK

ABSTRACT

It is widely known that significant in-sample evidence of predictability does not guarantee significant out-of-sample predictability. This is often interpreted as an indication that in-sample evidence is likely to be spurious and should be discounted. In this paper, we question this interpretation. Our analysis shows that neither data mining nor dynamic misspecification of the model under the null nor unmodelled structural change under the null are plausible explanations of the observed tendency of in-sample tests to reject the no-predictability null more often than out-of-sample tests. We provide an alternative explanation based on the higher power of in-sample tests of predictability in many situations. We conclude that results of in-sample tests of predictability will typically be more credible than results of out-of-sample tests.

Key Words: Predictive ability; Spurious inference; Data mining; Model instability.

*Correspondence: Atsushi Inoue, Department of Agricultural and Resource Economics, North Carolina State University, Box 8109, Raleigh, NC 27695-8109, USA; Fax: (919) 515-1824; E-mail: atsushi_inoue@ncsu.edu.

DOI: 10.1081/ETC-200040785 0747-4938 (Print); 1532-4168 (Online)
Copyright © 2004 by Marcel Dekker, Inc. www.dekker.com

ORDER REPRINTS

Mathematics Subject Classification: C22; C52; C53.

1. INTRODUCTION

A common problem in empirical work is that of assessing the predictability of one time series variable, given information about another variable. This problem involves a comparison of the predictive content of two nested forecast models subject to estimation uncertainty. The restricted model serves as the benchmark.[a] Our objective is to determine whether a linear predictive relationship exists in population. This objective is conceptually different from that of selecting the more accurate forecast model among the models in question, because in the latter case it may be advantageous to select a misspecified model (see Inoue and Kilian, 2003).

A leading example of the type of problem we have in mind are tests of the predictability of asset returns in empirical finance. For example, there is great interest in whether variables such as the dividend-price ratio or the earnings-price ratio help to predict future stock returns or excess returns (see Campbell and Shiller, 1988a,b; Fama and French, 1988; Goetzmann and Jorion 1993, 1995). Another example are tests for calendar effects in stock returns (see Sullivan et al., 2001). Predictability tests are also widely used in international finance. There is a large literature on testing the predictability of changes in the nominal exchange rate based on deviations of nominal exchange rates from macroeconomic fundamentals (see, e.g., Berkowitz and Giorgianni, 2001; Chinn and Meese, 1995; Faust et al., 2003; Kilian, 1999; Kilian and Taylor, 2003; Mark, 1995). Related studies have focused on the question of whether forward rates predict future spot rates (see Clarida and Taylor, 1997; Clarida et al., 2003). Meese and Rogoff (1983) and Meese and Rose (1991) have tested the predictability of the level of spot exchange rates based on the level of macroeconomic fundamentals. Mark and Choi (1997) and Meese and Rogoff (1988) test the predictability of real exchange rates. Other applications include tests of the predictive content of advertising for consumer spending (see Ashley et al., 1980), the predictive content of money for output (see Amato and Swanson, 2001), the predictive content of output-gap measures for inflation (see Clark, 2004), and the predictive content of asset prices for output and inflation (see Stock and Watson, 2003).

Predictability tests can be conducted based on the in-sample fit of a model or they can be based on the out-of-sample fit obtained from a sequence of recursive or rolling regressions. In the former case, we use the full sample in fitting the models of interest. Examples of in-sample tests are standard t-tests or F-tests. In the latter case we attempt to mimic the data constraints faced by a real-time forecaster. Examples of out-of-sample tests are tests of equal predictive accuracy and tests of forecast encompassing. Throughout this paper, we will focus on recursive out-of-sample tests for expository purposes.

If in-sample and out-of-sample tests of predictability tended to give the same answer, when applied to the same data set, it would not matter much, which one

[a]This paper does not deal with forecast accuracy tests for nonnested models (see, e.g., West, 1996).

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

we use. In practice, however, in-sample tests tend to reject the null hypothesis of no predictability more often than out-of-sample tests. It is important to understand why. One possible explanation that is widely accepted among applied researchers is that in-sample tests of predictability have a tendency to indicate spuriously the existence of predictability when there is none. In other words, the claim is that in-sample tests have a tendency to reject the no predictability null hypothesis more often than they should at the chosen significance level. If true, this explanation would suggest that we discount significant evidence in favor of predictability based on in-sample tests, if this evidence cannot also be supported by out-of-sample tests. For example, Ashley et al. (1980) insist that "a sound and natural approach" to testing predictive ability "must rely primarily on the out-of-sample forecasting performance" (p. 1149) and stress the likelihood of "spurious inferences ... when out of-sample verification is not employed" (p. 1165). The purpose of this paper is to question this interpretation.

First, we note that strong in-sample evidence and weak out-of-sample evidence are not necessarily an indication that in-sample tests are not reliable. We show that an alternative explanation of the same empirical regularity is that in-sample tests in many cases have higher power. We base this conclusion on a formal analysis of the asymptotic size and power properties of in-sample tests and out-of-sample tests of predictability.[b] If true, that alternative explanation would suggest that we discount the results out-of-sample tests, if the latter fail to confirm the findings of predictability using in-sample tests.

Second, we question the notion that in-sample tests of predictability are more susceptible to size distortions than out-of-sample tests. Why would in-sample tests generate spurious rejections of the no predictability null that persist asymptotically? One concern is the effect of data mining on the size of tests of predictability. Data mining is said to occur when a researcher searches over alternative forecast models, but only reports results for the specification with the highest predictive content (see, e.g., Granger, 1990). This practice will cause the size of the test of predictability to be inflated, resulting in spurious rejections of the no-predictability null and thus overfitting relative to the true model even asymptotically. Note that it is not necessary for any one researcher to mine the data deliberately. It suffices that several researchers independently consider alternative predictors and only significant results are ultimately published (see Denton, 1985).

If it were true that only in-sample tests are data-mined (or at least that in-sample tests are data mined more intensively than out-of-sample tests of predictability), then data mining would indeed provide a ready explanation of the tendency of in-sample tests to reject the null of no predictability more often than out-of-sample tests. It is a misperception, however, that out-of-sample tests are immune from data mining. The fundamental problem is that so-called out-of-sample tests are not truly "out-of-sample" in that the researcher is free to experiment with alternative predictors until he finds a significant predictor. Thus, data mining out-of-sample tests is no

[b]We focus on asymptotic results because finite-sample size distortions in practice can be effectively eliminated by the use of bootstrap methods (see, e.g., Clark and McCracken, 2004; Kilian, 1999; Kilian and Taylor, 2003; Mark, 1995; Rapach and Wohar, 2003).

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

more difficult than data mining in-sample tests. In fact, much of the recent literature on data mining has been directed exclusively at the problem of data mining in simulated out-of-sample inference (see, e.g., Hansen, 2001; White, 2000).

It is sometimes thought that as a practical matter there has been less data mining of out-of-sample tests than in-sample tests. There is no direct empirical evidence for this view, however. Individual data mining typically goes unreported by the perpetrator, and collective data mining is equally difficult to detect. Thus, we simply do not know that more data mining has taken place on in-sample tests than on out-of -sample tests. In fact, many authors report both in-sample and out-of-sample tests of predictability in the same paper, which argues against that presumption (see, e.g., Mark, 1995; Mark and Choi, 1997). Moreover, we cannot argue that the mere fact that in-sample tests reject the null more often indirectly proves the existence of data mining, because, as we will show in Sec. 2, the same pattern of rejection would be likely to occur even in the absence of data mining given the higher power of in-sample tests.

Thus, to the extent that data mining is a potential concern, we should be equally skeptical of in-sample and out-of-sample tests of predictability when standard critical values are used. A more constructive answer is possible in some cases. We show that under certain conditions the critical values of both in-sample and out-of-sample tests may be adjusted for data mining. Notably, when the set of potential predictors is finite and known, we can employ techniques in the spirit of White (2000) to construct conservative critical values under the presumption of systematic data mining across predictors. We derive such tests and compare the local asymptotic power of data mining robust out-of-sample tests relative to data-mining robust in-sample tests. We show that in-sample tests tend to have higher power than out-of-sample tests even after correcting for data mining.

An alternative and independent rationale for the apparently contradictory results of in-sample and out-of-sample tests of predictability is the presence of structural instability (see, e.g., Stock and Watson, 2003). We investigate the conjecture that in-sample tests are more prone than out-of-sample tests to generate spurious rejections when there is parameter instability due to unmodelled structural change. We distinguish between unmodelled structural breaks that result in a model that is misspecified under the null on the one hand and unmodelled structural breaks in the parameter of interest on the other. We show that the first type of structural break results in size distortions for both types of tests, without any presumption that these distortions are greater for in-sample tests of predictability, whereas the second type of break has no effect on the size of either test of predictability (although it may undermine the asymptotic power of out-of-sample tests, as shown by Clark and McCracken, 2003).

The power advantages of in-sample tests in the latter case hinge on the premise that the objective of predictability testing is to detect predictability, if it exists at all anywhere along the sample path. If instead the objective is to detect whether predictability holds at the end of the sample, then high power may be a disadvantage, as noted by Clark and McCracken (2003). Specifically, when a break *away from* predictability occurs in the sample, then out-of-sample tests because of their lower power will tend to provide a better indication of end-of-sample predictability. Clark and McCracken have suggested that such structural breaks may provide an alternative explanation of the empirical tendency of in-sample tests to reject more often

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

than out-of-sample tests. They also show that out-of-sample tests may have higher power in finite samples, provided that a break *toward* predictability occurs late in the sample. The latter result, of course, does not help to explain the empirical tendency of in-sample tests to reject more often than out-of-sample tests.

A third possible rationale for the tendency of in-sample tests of predictability to reject more often is the perception that in-sample tests are less robust than out-of-sample tests to dynamic misspecification under the null in the form of serially correlated regression errors. We compare t- and F-tests based on suitable robust standard errors to versions of out-of-sample tests of predictability that are robust to dynamic misspecification. We show that there is no presumption that in-sample tests are less reliable than out-of-sample tests under dynamic misspecification. All these tests are correctly sized asymptotically. Our analysis also shows that the robust out-of-sample tests in most cases have lower asymptotic power than robust in-sample tests. The power advantage of the robust t- and F-tests can be substantial.

The remainder of the paper is organized as follows. We study the asymptotic behavior of in-sample and out-of-sample tests under alternative assumptions. We begin in Sec. 2 with an environment that corresponds to the standard assumptions used in empirical work. This will be our benchmark. In Sec. 3, we consider an environment that is subject to systematic data mining along the lines described by Granger (1990), while abstracting from other complications. In Sec. 4, we will study an environment that allows for unmodelled structural change, but otherwise corresponds to the standard environment. Finally, in Sec. 5, we will focus on models that are dynamically misspecified under the null, while abstracting from structural change and data mining. We conclude in Sec. 6. The proofs are in the Appendix.

2. ANALYSIS UNDER STANDARD ASSUMPTIONS

In the environment without data mining or unmodelled structural change, standard critical values are adequate to prevent overfitting, and the choice between in-sample and out-of-sample tests of predictability reduces to the question of which test has higher power. We will therefore derive the asymptotic distributions of these tests under local alternatives in Sec. 2.1 and compare their asymptotic power in Sec. 2.2. Our power analysis will provide important insights that help to understand the tendency of in-sample tests of predictability to reject more often than out-of-sample tests.

2.1. The Asymptotic Distributions of Tests of Predictability Under Local Alternatives

Consider

$$y_{t+1} = \gamma' x_t + u_{t+1} = \alpha' v_t + \beta' w_t + u_{t+1}, \tag{1}$$

where $x_t = (v_t', w_t')'$, v_t and w_t are m, l and k dimensional vectors of regressors, respectively, $\{u_{t+1}\}$ is a sequence of martingale differences with respect to the sigma

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

field generated by $\{x_t, x_{t-1}, x_{t-2}, \ldots\}$. The m.d.s. assumption arises naturally in many applications of predictability tests. An example are tests of the random walk hypothesis for nominal exchange rates (see Mark, 1995). Equation (1) applies to predictions one period ahead, but is readily modified for multi-step ahead predictions. For example, we may be interested in predicting the change in the exchange rate over the next four quarters. Because the data are overlapping in that case, the error term will be serially correlated and no longer a m.d.s. as assumed here. We will discuss this case in Sec. 5.

Definition 1 (Predictability). y_{t+1} is said to be *predictable* based on w_t if and only if $\beta \neq 0$.

Assumption 1.

(a) $\{(x_t' u_{t+1}, \mathrm{vech}(x_t x_t')')\}$ is covariance stationary.
(b) $\{x_t x_t'\}$ satisfies the uniform law of large numbers: $(1/T)\sum_{t=1}^{[Tr]} x_t x_t' \xrightarrow{p} rE(x_0 x_0')$ where the convergence is uniform in r and $E(x_0 x_0')$ is positive definite.
(c) $\{x_t u_{t+1}\}$ satisfies the functional central limit theorem (FCLT): $T^{-1/2}\sum_{t=1}^{[Tr]} x_t u_{t+1} \Rightarrow \Sigma^{1/2} W(r)$.
(d) There is a consistent estimator of Σ, $\widehat{\Sigma}$. That is, $\widehat{\Sigma} \xrightarrow{p} \Sigma$.

Assumption 2. $\Sigma = \sigma^2 E(x_0 x_0')$ for some $\sigma^2 > 0$.

Assumption 1(d) rules out some important and interesting cases, such as the near-cointegrated case analyzed by Rossi (2003) (also see Corradi et al., 2001). Assumption 2 imposes conditional homoskedasticity. This is just for notational convenience and could be relaxed.

We are interested in testing

$$H_0 : \beta = 0_{k\times 1}$$

against the two-sided alternative

$$H_1 : \beta \neq 0_{k\times 1}.$$

For some applications one is interested in testing $H_0 : \beta_1 = 0_{1\times 1}$ against a more specific one-sided alternative

$$H_2 : \beta_1 > 0_{1\times 1}.$$

with $k = 1$. Both H_1 and H_2 are potentially relevant for empirical work, but, in practice, H_2 is often the economically more interesting alternative hypothesis. Tests of predictability are typically motivated by the fact that economic models imply the existence of a linear predictive relationship in the data. The same economic models, however, also typically tell us in what direction we should see a departure from the null of no predictability if the economic model is true. For example, the monetary model of exchange rate determination implies that the exchange rate should fall if

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

today's exchange rate exceeds its fundamental value. For that reason, every study of this model has relied on one-sided t-tests (see, e.g., Mark, 1995). The same logic applies to testing the predictability of the dividend-ratio model of stock returns (or similar models based on the earnings-price ratio). Similarly, in Mark and Choi's (1997) study of the predictability of real exchange rates higher growth or higher government spending lead to a real appreciation, so the natural alternative is the one-sided alternative. In contrast, two-sided alternatives are primarily relevant for models with $k > 1$.

For each alternative hypothesis, we consider a sequence of local alternatives. That is,

$$H_1' : \beta = T^{-1/2}c$$

and

$$H_2' : \beta_1 = T^{-1/2}c_1,$$

respectively, where c is a k-dimensional nonzero vector and c_1 is a positive number.

We begin by defining the in-sample test statistics. For testing H_0 against H_1, we consider the test statistic

$$F = \frac{\sum_{t=1}^{T}(\hat{u}_{0t+1}^2 - \hat{u}_{1t+1}^2)}{\hat{\sigma}^2} \tag{2}$$

where $\{\hat{u}_{1t+1}\}$ are the unrestricted OLS residuals, $\{\hat{u}_{0t+1}\}$ are the restricted OLS residuals under H_0 and $\hat{\sigma}^2 = (1/T)\sum_{t=1}^{T}\hat{u}_{1t+1}^2$. For testing H_0 against H_2, we consider the test statistic

$$t = \sqrt{T}\hat{\beta}_1/\hat{\sigma}_{\beta_1} \tag{3}$$

where $\hat{\sigma}_{\beta_1}^2$ is the $(l+1)\times(l+1)$ element of $\hat{\sigma}^2((1/T)\sum_{t=1}^{T}x_t x_t')^{-1}$. The following proposition provides the asymptotic distributions of F and t under a sequence of local alternatives.

Proposition 1. *Suppose that Assumption* 1 *holds. Under a sequence of local alternatives of the form* H_1' *and* H_2', *respectively,*

$$F \xrightarrow{d} \chi_{\delta_1}^2(k) \tag{4}$$

$$t \xrightarrow{d} N(\delta_2, 1), \tag{5}$$

where $\chi_{c_1}^2(k)$ *is a noncentral* χ^2 *random variable with degree of freedom* k *and noncentral parameters*

$$\delta_1 = \frac{1}{\sigma^2}c'\{E(w_t w_t') - E(w_t v_t')[E(v_t v_t')]^{-1}E(v_t w_t')\}c$$

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

and

$$\delta_2 = e'_{l+1}(1/\sigma_{\beta_1})[E(x_t x'_t)]^{-1} E(x_t w'_{t1}) c_1,$$

where e_{l+1} is an $(l+1)$-dimensional column vector, whose last element is one and other elements are all zero, $\sigma^2_{\beta_1}$ is the $(l+1)\times(l+1)$ element of $\sigma^2[E(x_t x'_t)]^{-1}$, and w_{t1} is the first element of w_t.

The proof of Proposition 1 is straightforward and thus is omitted. The null limit distribution emerges as a special case of this result with $c = \mathbf{0}$ and $c_1 = 0$, respectively.

Next, we define the recursive out-of-sample test statistics. We fit the model by OLS on the first S observations and evaluate the fit (loss) on observation $S+1$, for $S = R, R+1, R+2, \ldots, T-1$. The recursive OLS estimators are defined by $\hat{\gamma}_S = [\hat{\alpha}'_S, \hat{\beta}'_S]' = (\sum_{s=1}^{S} x_s x'_s)^{-1} \sum_{s=1}^{S} x_s y_s$, and $\tilde{\alpha}_S = (\sum_{s=t}^{S} v_s v'_s)^{-1} \sum_{s=1}^{S} v_s y_s$ under H_0. We consider the recursive F statistic:

$$GM = \frac{\sum_{t=R+1}^{T} (\bar{u}^2_{0t+1} - \bar{u}^2_{1t+1})}{\hat{\sigma}^2} \tag{6}$$

where $\bar{u}_{0t+1} = y_{t+1} - \tilde{\alpha}'_{t-1} v_t$, and $\bar{u}_{1t+1} = y_{t+1} - \hat{\gamma}'_{t-1} x_t$. McCracken (1999) calls (6) the modified OOS-F statistics. Gilbert (2001) considers the same statistic in the split-sample context. Thus we will refer to (6) as the Gilbert–McCracken (GM) test statistic. The GM-test statistic is similar to the DM-test statistic of Diebold and Mariano (1995) in that it is based on the average loss differential. The key difference is that the GM-test is designed for comparing nested models, whereas the DM-test is designed for comparing nonnested models. The DM-test is designed for models that are not subject to parameter estimation error, but with a different choice of covariance estimator it also may be used for nonnested models with estimation uncertainty. The latter test has been proposed by West (1996). The problem with using West's test in the nested case with estimation uncertainty is that the test statistic vanishes in probability rather than converging to a normal limiting distribution. The GM-test addresses this problem by replacing the denominator of the test statistic by $\hat{\sigma}^2$.

Assumption 3. $R/T \to \pi \in (0,1)$ as $T \to \infty$.

Proposition 2. *Suppose that Assumptions* 1, 2 *and* 3 *hold. Under a sequence of local alternatives of the form H'_1*

$$GM \xrightarrow{d} \int_\pi^1 \frac{2}{r} W'(r) dW(r) + \delta' \int_\pi^1 \frac{2}{r} W(r) dr + 2\delta'(W(1) - W(\pi)) + 2(1-\pi)\delta'\delta$$
$$- \int_\pi^1 \frac{1}{r^2} (W(r) + r\delta)'(W(r) + r\delta) dr \tag{7}$$

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

where $W(\cdot)$ is a k-dimensional standard Brownian motion, $\delta = (1/\sigma)L'E(x_t x_t')^{-1/2} E(x_t w_t')c$, L is a $(k+l) \times k$ matrix that satisfies $LL' = Q$ and $L'L = I_k$, and

$$Q = [E(x_t x_t')]^{\frac{1}{2}} \left\{ (E(x_t x_t'))^{-1} - \begin{bmatrix} (E(v_t v_t'))^{-1} & 0_{l\times k} \\ 0_{k\times l} & 0_{k\times k} \end{bmatrix} \right\} [E(x_t x_t')]^{\frac{1}{2}}.$$

The result in (7) is due to McCracken (1999, Theorem 4.1). The null limit distribution is a special case of (7) with $c = \mathbf{0}$.

An alternative approach to comparing the forecast accuracy of two models is the encompassing test, which involves running the regressions:

$$y_{t+1} = \alpha' v_t + u_{0t+1}$$
$$y_{t+1} = \alpha' v_t + \beta' w_t + u_{1t+1}$$

and testing the null hypothesis

$$E(u_{0t+1}^2) - E(u_{0t+1} u_{1t+1}) = 0$$

against the alternative $E(u_{0t+1}^2) - E(u_{0t+1} u_{1t+1}) > 0$. Clark and McCracken (2001) consider the encompassing test

$$ENC = \frac{(T-R)\sum_{t=R+1}^{T} (\bar{u}_{0t+1}^2 - \bar{u}_{0t+1}\bar{u}_{1t+1})}{\sum_{t=R+1}^{T} \bar{u}_{1t+1}^2}. \tag{8}$$

They show that this F-type test is more powerful than t-type tests of forecast encompassing. Their test for nested forecast models differ from the test in Harvey et al. (1998) in that it allows for parameter estimation uncertainty. West (2001) considers encompassing tests for nonnested models with estimation uncertainty.

Proposition 3. *Suppose that Assumptions* 1, 2 *and* 3 *hold. Under a sequence of local alternatives of the form H_1'*

$$ENC \xrightarrow{d} \left[\int_\pi^1 \frac{1}{r} W'(r) dW(r) + \delta' \int_\pi^1 \frac{1}{r} W(r) dr + \delta'(W(1) - W(\pi)) + (1-\pi)\delta'\delta \right]. \tag{9}$$

The null limit distribution of this encompassing test has been derived by Clark and McCracken (2001, *Theorem* 3.1). *This null limit distribution is obtained from* (9) *by setting $c = \mathbf{0}$.*

2.2. Asymptotic Power Comparison of In-Sample and Out-of-Sample Tests of Predictability

We evaluate the local asymptotic power of the four predictability tests of Sec. 2.1 by simulation. We focus on a stylized example that is similar to processes studied in

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

the empirical literature. Let $l = 1$ and $k = 1$, corresponding to the standard specification used for example in testing the dividend-ratio model. Thus, F may be interpreted as a two-sided t-test and t as a one-sided t-test of the null hypothesis that w_t does not predict $y_{(t+1)}$. We postulate that $v_t = 1$ for all t, and $w_t = \rho w_{t-1} + \eta_t$. For simplicity, let $\eta_t \sim NID(0, \sigma_\eta^2)$ and $u_{t+1} \sim NID(0, \sigma^2)$. We set $\rho = 0.9$, $\sigma_\eta^2 = 0.005$ and $\sigma^2 = 0.05$. These values are close to values obtained in empirical research (see, e.g., Mark, 1995). Our qualitative conclusions are not sensitive to these parameter choices. Specifically, we evaluated a grid of parameter values including $\rho \epsilon \{0.45, 0.9, 0.99\}$, $\sigma_\eta^2 \epsilon \{0.005, 0.05\}$ and $\sigma^2 \epsilon \{0.05, 0.5\}$. The results remained very similar.

We set $\pi \epsilon \{0.3, 0.4, 0.5, 0.6, 0.7\}$ and $c \epsilon \{0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5\}$. The larger c, the larger the power of the test, all else equal. For $c = 5$ the power of the in-sample tests is close to 100%. We evaluate the local asymptotic power of all tests at the nominal 10% significance level. The asymptotic critical values are computed based on 20000 draws from the limit distributions in Sec. 2 with $c = 0$. The rejection rates under the alternative are based on 20000 draws from the limit distribution with $c > 0$. All Brownian motions in turn are approximated based on discrete time approximations with $T = 5000$.

In Fig. 1, the asymptotic power of the out-of-sample tests GM and ENC is expressed in ratio form relative to the asymptotic power of the in-sample F-test (upper panel) and in-sample t-test (lower panel). Power ratios below 1 indicate that the in-sample test is more powerful. We find that for all values of π and c, the t-test

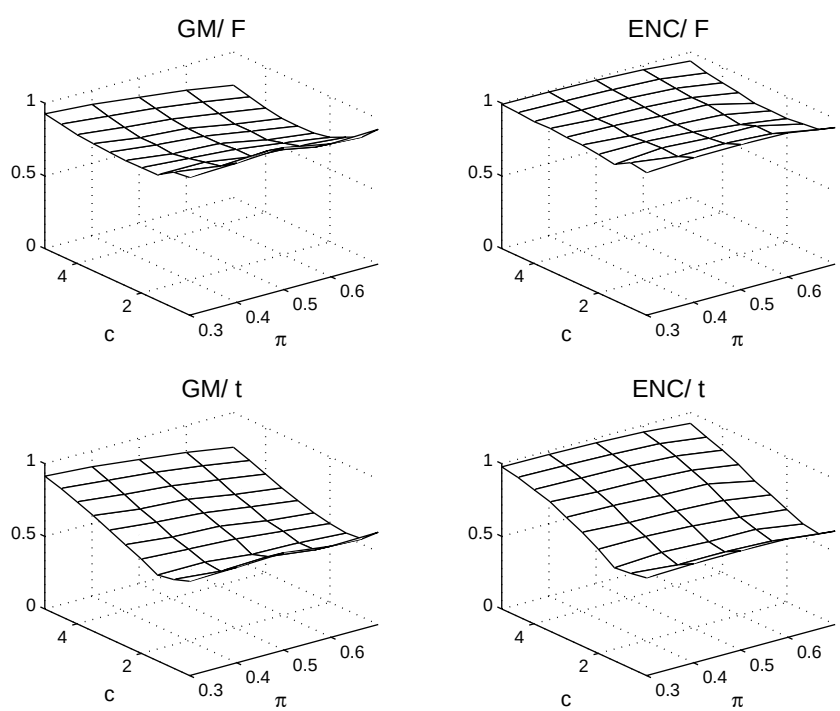

Figure 1. Relative power of predictability tests in standard environment.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

and the F-test have higher power than the GM-test as well as higher power than the ENC-test. Thus, if a researcher is interested in testing H_2 our results suggest that the preferred testing strategy, at least asymptotically, would be to rely on the one-sided t-test. Similarly, if we are interested in the two-sided alternative H_1, the F-test would be the preferred test.

The existence of power advantages for the in-sample tests is not surprising in this standard setting. When the conditions of the Neyman–Pearson lemma are satisfied, one would expect in-sample tests to be the locally asymptotically most powerful tests. Nevertheless, the results in Fig. 1 provide useful additional insights. First, they give us an indication of how large the power differences are. Although in some cases the out-of-sample tests reach between 96% and 98% of the power of the in-sample tests, there are others where the out-of-sample tests have much lower power. For example, the GM-test may have power as low as 53% of the power of the t-test and 70% of the power of the F-test, even asymptotically. The ENC-test may have power as low as 85% of the power of the F-test and 51% of the power of the t-test. Thus, the differences are potentially large enough to provide an alternative explanation of the observed tendency of in-sample tests to reject the null of no predictability more often than out-of-sample tests, *even* when no data mining has taken place and when there is no structural instability or dynamic misspecification under the null.

Second, it is widely believed that–if there is data mining at all on the out-of-sample test – the extent of data mining is bound to be greater for the in-sample test. There is of course no direct evidence for that assertion, but this position is sometimes defended by the following argument: Given that so many published studies report out-of-sample failures of a model that fits well in-sample, it is asserted that the data mining problem cannot be very prevalent for the out-of-sample test. After all it would seem that if both tests were mined to the same extent, they both would tend to reject. Figure 1 shows that this view is mistaken. Even if (hypothetically) the same extent of data mining has taken place on both tests, the potentially much lower power of the out-of-sample test may offset the increased size induced by data mining. As a result, one will tend to observe more rejections for the in-sample test.

Third, the relative power advantages of the t-test are even more pronounced than for the F-test. This result is not surprising because the t-test is a test of the null of no predictability against the one-sided alternative H_2. In contrast, the out-of-sample tests GM and ENC test the weaker hypothesis of whether one model is a more accurate predictor than the other, which corresponds to H_1. Thus, their power is diluted. There are no out-of-sample tests, to our knowledge, that can incorporate one-sided hypotheses on β implied by economic theory. For example, standard exchange rate models imply that periods, in which the nominal exchange rate exceeds the equilibrium value of the exchange rate, should be followed by a *depreciation* of the exchange rate, as opposed to a *change* in the exchange rate. In other words, the economic model pins down the direction of change. Out-of-sample tests do not make use of that restriction. It immediately follows that one cannot use the out-of-sample tests to ''verify'' the test results obtained using the one-sided t-test. This point does not appear to have been made in the literature, but helps to understand the observed differences in test results.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

3. ANALYSIS IN THE PRESENCE OF DATA MINING

It is well understood that under standard assumptions neither in-sample tests nor out-of-sample tests of predictability will have a tendency to generate spurious rejections of the null. One candidate explanation for the alleged tendency of in-sample tests to reject the null hypothesis of no predictability too often is the practice of data mining (see, e.g., Granger, 1990). Standard critical values are constructed under the presumption that no data mining has taken place. If we use these critical values for in-sample inference after data mining has taken place, then in-sample tests will reject the null too often. Of course, an analogous size distortion would arise if the out-of-sample tests were data-mined, so a crucial part of the argument is that data mining is less prevalent in out-of-sample inference than in in-sample inference. We will argue that this premise is not very plausible.

For example, it is sometimes argued that it is more difficult to data mine out-of-sample tests than in-sample tests. This position is hard to defend. In practice, out-of-sample tests are used not unlike cross-validation methods. The researcher uses only the first part of the T observations in the sample for specification searches and sets aside the second part of the data for the evaluation of the preferred model. This sample splitting strategy is often thought to prevent data mining at the individual level. It is clear, however, that this strategy is not immune from data mining. Nothing ensures that the researcher who presents pseudo out-of-sample validation results in his paper has not experimented with other predictors without showing the results. The problem is that, when the simulated out-of-sample exercise is completed, the researcher knows exactly the ''out-of-sample'' -performance of any given model and is free to experiment with alternative predictors prior to publication. Even if the researcher has only considered one predictor, there remains the danger of collective data mining (see Denton, 1985). Thus, simulated out-of-sample inference is subject to exactly the same potential data mining problems as in-sample inference.[c] In fact, much of the recent literature on data mining has explicitly focused on the problem of data mining in simulated out-of-sample inference (see, e.g., Hansen, 2001; White, 2000). It may seem that one could address this problem by evaluating studies only several years after publication, as new data have become available. This, of course, is not the way these tests are currently used, and even if one did use them in this fashion, this would not eliminate the problem of collective data mining on the ex post evaluation results.

An alternative view recognizes the scope for data mining out-of-sample test results, but contends that, in practice, more data mining takes place on in-sample tests than on out-of-sample tests of predictability. That position lacks solid empirical support. There is no direct evidence on the extent of data mining because data mining by design can rarely be detected, so we really do not know that data mining is empirically more of a problem for in-sample tests. It may be tempting to argue that

[c]McCracken (2001) studies out-of-sample inference involving forecasting models that in turn were selected based on some inconsistent model selection procedure. His methodology, however, presumes that no respecification of the forecast model occurs after the out-of-sample test is conducted. Thus, he rules out data mining of the form described here.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

the mere fact that in-sample tests tend to reject more often is indirect evidence of disproportionate in-sample data mining, but, as we have shown in Sec. 2, we would expect that tendency even in the absence of data mining because of the higher power of in-sample tests. In fact, there is reason to believe that the concern over in-sample data mining is overrated. Many empirical studies of predictability report results for in-sample and out-of-sample tests side by side, which argues against the presumption that more data mining has taken place on the in-sample test (see e.g., Kilian, 1999; Mark, 1995; Mark and Choi, 1997; Rapach and Wohar, 2003). To the extent that there is scope for data mining, however, we need to recognize that even the results of out-of-sample tests of predictability are likely to be contaminated by data mining at the individual or the collective level. There is no a priori presumption that the size distortions induced by data mining are greater for in-sample tests than for out-of-sample tests.

Given that both in-sample tests and out-of-sample tests of predictability, as currently used, will be rendered unreliable by data mining, we conclude that neither test can be recommended for applied work when potential data mining is a concern. A more constructive response is possible to the extent that the set of potential predictors is finite and known. This assumption will not be reasonable in all cases, but seems plausible for some important empirical applications such as the literature on exchange rate predictability. In that case, we can adjust the critical values of both in-sample and out-of-sample tests to account for data mining. Conservative critical values may be constructed by pretending that researchers have systematically searched for the test result most favorable to the rejection of the null. This proposal is in the spirit of recent work by White (2000) and by Hansen (2001), who proposed bootstrap methods for out-of-sample inference in the presence of data mining.[d] Since White's theoretical results presume that the out-of-sample test statistic of interest has a Gaussian limit distribution, they cannot be applied to the test statistics of interest in our paper. In Sec. 3.1, we propose a natural generalization of White's approach and derive the appropriate limit distributions for our out-of-sample tests. This allows us to construct critical values that are robust to data mining. We also derive the limit distribution of the in-sample tests of predictability under the assumption that systematic data mining has taken place. This allows us to construct appropriate data-mining robust critical values for the in-sample t- and F-tests.[e] Clearly, both in-sample and out-of-sample tests of predictability based on these robust critical values will be free of size distortions asymptotically. Thus, the choice between in-sample and out-of-sample inference again reduces to a question about relative power. We derive the asymptotic distributions of these tests under local alternatives in Sec. 3.1 and compare their asymptotic power in Sec. 3.2.

[d]Hansen (2001) discusses some possible drawbacks of White's proposal. Note that these possible drawbacks do not apply in our context because our model is nested and the null hypothesis holds with equality.

[e]Our analysis is a natural extension of work in classical statistics on the testing of multiple hypotheses (see, e.g., Anderson, 1994; Dasgupta and Spurrier, 1997; Royen, 1984). A similar framework has also been used by Hansen (2000) who proposed bootstrap inference for the distribution of R^2 in the presence of data mining.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

3.1. The Asymptotic Distributions of Data-Mining Robust Tests of Predictability Under Local Alternatives

The presence of data mining affects the distribution of in-sample and out-of-sample tests of predictability both under the null and under the alternative. In this section we derive suitable critical values for predictability tests that account for data mining and we derive their local asymptotic power. We formalize data mining as follows. For $j = 1, 2, \ldots, M$, let $w_{j,t}$ denote a k_j-dimensional subvector of w_t where $1 \leq k_j \leq k$. Let $x_{j,t} = [v_t' \; w_{j,t}']'$. M is finite and w_t is known. Define Q_j and L_j as in Proposition 2 with x_t replaced by $x_{j,t}$ and k replaced by k_j. Suppose that one is interested in whether any one of M models

$$y_{t+1} = \alpha' v_t + \beta_j' w_{j,t} + u_{t+1} \quad j = 1, 2, \ldots, M$$

has predictive power that is superior to the benchmark model

$$y_{t+1} = \alpha' v_t + u_{t+1}.$$

Formally, the null hypothesis can be written as

$$H_0 : \beta_j = 0 \quad \forall j$$

and the alternative hypothesis as

$$H_1 : \beta_j \neq 0 \quad \text{for some } j.$$

Under the null hypothesis, we have

$$\max_{j \in \{1,\ldots,M\}} E(u_{0,t+1}^2) - E(u_{j,t+1}^2) = 0,$$

whereas under the alternative hypothesis we have

$$\max_{j \in \{1,\ldots,M\}} E(u_{0,t+1}^2) - E(u_{j,t+1}^2) > 0,$$

where $u_{0,t+1} \equiv y_{t+1} - \alpha' v_t$ and $u_{j,t+1} \equiv y_{t+1} - \alpha' v_t - \beta_j' w_{j,t}$ for $j = 1, 2, \ldots, M$. This suggests the following in-sample test statistic:

$$F^{\max} = \max_{j \in \{1,\ldots,M\}} \frac{\sum_{t=1}^{T} (\hat{u}_{0,t+1}^2 - \hat{u}_{j,t+1}^2)}{\hat{\sigma}_j^2}. \tag{10}$$

The statistic $F^{\max}$ is the data-mining proof version of F.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

Let

$$\Omega = \begin{bmatrix} \Omega_1 1 & \cdots & \Omega_{M1} \\ \vdots & \Omega_{ij} & \vdots \\ \Omega_{1M} & \cdots & \Omega_{MM} \end{bmatrix},$$

$$d = [d_1 \ d_2 \ \cdots \ d_M]',$$

where

$$\Omega_{ij} = L_i' E(x_{i,t} x_{i,t}')^{-1/2} E(x_{i,t} x_{j,t}) E(x_{j,t} x_{j,t}')^{-1/2} L_j,$$
$$d_j = (1/\sigma) L_j' [E(x_{j,t} x_{j,t}')]^{-1/2} E(x_{j,t} w_{j,t}') c(\zeta_j),$$

σ^2 is the variance of u_{t+1}, and ζ_j denotes a k-dimensional selection vector whose jth element is 1 if $w_{j,t}$ includes the jth element of w_t and is zero otherwise. The notation $c(\zeta)$ stands for a subset of c where the selection vector $\zeta \in \{0,1\}^k$ and $1 \le k_j = \zeta'\zeta \le k$. For example, if $\zeta = [0,1,0,1,0]$ then $c(\zeta) = [c_2, c_4]'$.

Assumption 4. Ω is positive definite.

Proposition 4. *Suppose that Assumptions* 1, 2 *and* 4 *hold. Under a sequence of local alternatives of the form* H_1',

$$F^{\max} \xrightarrow{d} \max_{j \in \{1,\ldots,M\}} \chi^2_{d_j}(k_j) \tag{11}$$

where

$$\chi^2_{d_1}(k_1) = \sum_{i=1}^{k_1} u_i^2, \chi^2_{d_j}(k_j) = \sum_{i=k_1+\cdots+k_{j-1}+1}^{k_1+\cdots+k_j} u_i^2 \quad \textit{for } j = 2, \ldots, M,$$

and

$$\begin{bmatrix} u_1 \\ \vdots \\ u_{k_1+\cdots+k_M} \end{bmatrix} \sim N(d, \Omega).$$

The distribution of $[\chi^2_{d_1}(k_1), \chi^2_{d_2}(k_2), \ldots, \chi^2_{d_M}(k_M)]'$ is called a noncentral multivariate χ^2 distribution in the statistical literature (see Royen, 1997, for a recent survey).

Next, consider the special case in which $w_{j,t}$ is the jth element of w_t. The alternative hypothesis of interest is

$$H_2 : \beta_j > 0 \quad \text{for some } j,$$

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

and the sequence of local alternatives is

$$H_2' : \beta_j = T^{-1/2} c_j \quad j = 1, 2, \ldots, M,$$

where $c_j \geq 0$ for all j and $c_j > 0$ for at least one j. For H_2 it is natural to consider the following in-sample test

$$t^{\max} = \max_{j \in \{1,\ldots,M\}} \sqrt{T} \hat{\beta}_j / \hat{\sigma}_{\beta_j} \tag{12}$$

where $\hat{\sigma}^2_{\beta_j}$ is a consistent estimator of the asymptotic variance of $\widehat{\beta}_j$. S_8 is the data-mining-proof version of S_2. Let Σ be an $M \times M$ matrix whose (i, j)th element is given by

$$\sigma_{ij} = \sigma^2 e'_{l+1} [E(x_{i,t} x'_{i,t})]^{-1} E(x_{i,t} x_{j,t}) [E(x_{j,t} x'_{j,t})]^{-1} e_{l+1} / \sigma_{\beta_i} \sigma_{\beta_j}$$

and let f be an M-dimensional column vector whose jth element is given by

$$f_j = e'_{l+1} [E(x_{j,t} x'_{j,t})]^{-1} E(x_{j,t} w_{j,t}) c_j / \sigma_{\beta_j},$$

where e_{l+1} is an $(l+1)$-dimensional column vector whose last element is one and other elements are all zero.

Assumption 5. Σ is positive definite.

Proposition 5. *Suppose that Assumptions* 1, 2 *and* 5 *hold. Under a sequence of local alternatives of the form* H_2'

$$t^{\max} \xrightarrow{d} \max_{j \in \{1,\ldots,M\}} (v_j) \tag{13}$$

where

$$\begin{bmatrix} v_1 \\ \vdots \\ v_M \end{bmatrix} \sim N(f, \Sigma).$$

For $c = \mathbf{0}$, (11) and (13) include the null distribution as a special case. Thus, we are able to derive critical values that are robust to data mining. Note that a key assumption underlying this strategy is that the regression models in question are linear. Under nonlinearity the number of models that could be constructed from a

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

given set of predictors would be infinite and the proposed strategy for computing data-mining robust critical values would be infeasible.

Next, we consider the data-mining robust out-of-sample tests. Define the maximum versions of the recursive GM test and the recursive encompassing test as follows:

$$GM^{\max} = \max_{j\in\{1,\ldots,M\}} \frac{\sum_{t=R+1}^{T}(\bar{u}_{0t+1}^2 - \bar{u}_{j,t+1}^2)}{\hat{\sigma}_j^2},$$
$$ENC^{\max} = \max_{j\in\{1,\ldots,M\}} \frac{(T-R)\sum_{t=R+1}^{T}(\bar{u}_{0t+1}^2 - \bar{u}_{0t+1}\bar{u}_{j,t+1})}{\sum_{t=R+1}^{T}\bar{u}_{jt+1}^2}. \tag{14}$$

Proposition 6. *Suppose that Assumptions* 1, 2, 3 *and* 4 *hold. Then under a sequence of local alternatives of the form* H_1'

$$\begin{aligned} GM^{\max} \xrightarrow{d} \max_{j\in\{1,\ldots,M\}} & \int_\pi^1 \frac{2}{r} B_j'(r)dB_j(r) \\ & + 2d_j{}' \int_\pi^1 \frac{1}{r} B_j(r)dr + 2d_j'(B_j(1) - B_j(\pi)) + 2(1-\pi)d_j'd_j \\ & - \int_\pi^1 \frac{1}{r^2}(B(r) + rd_j)'(B_j(r) + rd_j)dr, \end{aligned} \tag{15}$$

$$\begin{aligned} ENC^{\max} \xrightarrow{d} \max_{j\in\{1,\ldots,M\}} \Bigg[& \int_\pi^1 \frac{1}{r} B_j'(r)dB_j(r) + d_j' \int_\pi^1 \frac{1}{r} B_j(r)dr \\ & + d_j'(B_j(1) - B_j(\pi)) + (1-\pi)d_j'd_j \Bigg], \end{aligned} \tag{16}$$

where $\{B_j(\cdot)\}_{j=1}^M$ *are Brownian motions that satisfy* $E(B_i(r)B_j(s)') = \min(r,s)\Omega_{ij}$.

By setting $c = \mathbf{0}$, (15) and (16) include the null limit distribution as a special case.

One complication in implementing our approach in practice is that the limit distribution will in general depend on the set of potential predictors and their time series properties. Although we rarely know the search path chosen by other researchers, in many cases it seems reasonable to presume that individually or collectively researchers will explore the universe of economically plausible predictors that is publicly available. For example, in the context of quarterly foreign exchange returns it seems reasonable to include all quarterly macro variables that have been used in previous studies as potential exchange rate predictors. The set of these variables has remained remarkably stable over time. It is straightforward to approximate the time series processes of these macro variables by bootstrap methods and to construct data-mining critical values under the null hypothesis. Indeed a bootstrap version of our tests has recently been implemented by Rapach and Wohar (2002) in the context of testing the predictability of annual and quarterly stock returns.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

3.2. Asymptotic Power Comparison of In-Sample and Out-of-Sample Tests of Predictability

We now turn to the asymptotic power comparison for environments in which more than one candidate model was considered prior to testing. The economically most interesting alternative hypothesis is one, in which one candidate model helps to predicts y_{t+1} in population, whereas all other candidate models do not. For expository purposes we let $M = 2$. Thus, $c = [c_1 \;\; 0]$, where $c_1 \epsilon \{0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5\}$. We postulate that the two potential predictors w_{jt} follow identical AR(1) processes that are independent. The details of the design are otherwise identical to the environment without data mining. In short, the experiment differs from the previous subsection in that we allow for the possible selection of an irrelevant predictor.

In Fig. 2, the asymptotic power of the out-of-sample tests $GM^{\max}$and $ENC^{\max}$ is expressed in ratio form relative to the asymptotic power of the in-sample tests $F^{\max}$ (upper panel) and $t^{\max}$ (lower panel). As in the case without data mining, both the one-sided $t^{\max}$-test and the $F^{\max}$-test have higher power than the out-of-sample tests for all design points we considered. The magnitude of the power advantage of the in-sample test can be substantial. In some cases, the $GM^{\max}$-test has only 69% of the power of the $F^{\max}$-test and the $ENC^{\max}$-test only 84%. Relative to the $t^{\max}$-test, these ratios may drop as low as 56% and 63%, respectively. It is important to note that

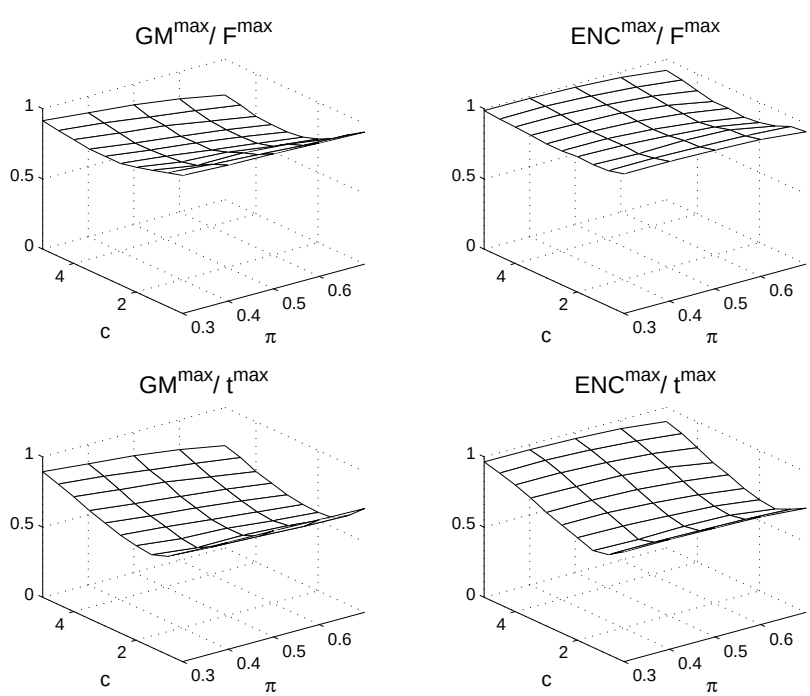

Figure 2. Relative power of predictability tests that are robust against data mining.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

these results were not ex ante obvious. Whereas the qualitative results in Fig. 1 may be expected based on the Neyman–Pearson lemma, this is not the case for the results in Fig. 2 because data mining makes the problem nonstandard. Nevertheless, the asymptotic power results are qualitatively similar to the standard case.

4. ANALYSIS IN THE PRESENCE OF UNMODELLED STRUCTURAL CHANGE

A possible alternative rationale for the apparently contradictory results of in-sample and out-of-sample tests of predictability is the presence of structural instability (see, e.g., Stock and Watson, 1999, p. 298, 326; Stock and Watson, 2003). So far, we have maintained the assumption that the process in question is not subject to unmodelled parameter instability. We now relax that assumption. For expository purposes, we will abstract from data mining.

Recall that

$$y_{t+1} = \gamma' x_t + u_{t+1} = \alpha' v_t + \beta' w_t + u_{t+1}, \tag{17}$$

where $x_t = (v_t', w_t')'$, v_t and w_t are m, l and k dimensional vectors of regressors, respectively, and $\{u_{t+1}\}$ is a sequence of martingale differences. There are two possibilities for unmodelled parameter instability to affect the result. For concreteness, we focus on the possibility of a one-time change in the parameter values. One possibility is that there is a break in the parameter of interest, β. The other possibility is that the structural change occurs in the conditioning set, i.e., in the parameter α. The latter case is an example of a model that is misspecified under the null hypothesis. It turns out that the consequences of not modelling such a break depend crucially on whether the break occurs in α or in β.

4.1. Structural Break in β

What is the effect of such a break on the size of predictability tests? Under the null hypothesis of no predictability we have that $\beta_t = 0\ \forall t$. This means that by definition of the null a structural break in the predictive relationship of interest is not possible. If $\beta_t \neq 0$ for some t, then clearly there is some predictability and the null of no predictability cannot hold. It follows immediately that our earlier results on the asymptotic size of in-sample and out-of-sample tests of predictability continue to apply. This point is important because it implies that deterministic structural breaks in the predictive relationship could not possibly be responsible for the alleged tendency of in-sample tests to result in spurious predictability.

Unmodelled structural breaks in β, however, may affect the power of tests of predictability. Clark and McCracken (2003) recently have studied the effects of deterministic structural breaks on the power of predictability tests. Their analysis abstracts from data mining. Like our paper they consider two parametric linear models that are nested under the null. The joint null hypothesis is no predictability and no structural change. Clark and McCracken allow for a one-time structural shift

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

in the predictive relationship under the alternative hypothesis. They postulate that for some part of the sample $\beta_t = 0$, whereas for the remainder of the sample $\beta_t \neq 0$. Under these conditions, Clark and McCracken (2003) show that the usual in-sample F-test will detect predictability with probability 1 asymptotically, if it exists at all anywhere along the sample path. In contrast, out-of-sample tests such as tests of equal forecast accuracy or forecast encompassing may be inconsistent against at least some structural break alternatives. Thus, if there is predictability at least for some part of the sample, out-of-sample tests may fail to detect it, even as in-sample tests correctly reject the no predictability null.[f]

Thus structural change under the alternative may provide an independent explanation of the empirical tendency of in-sample tests to reject the null more often than out-of-sample tests (see Clark and McCracken, 2003 for some evidence in this regard). It also provides an additional argument for attaching more credibility to the results of in-sample tests provided the objective is to detect predictability, if it exists at all anywhere along the sample path. If instead the objective is to detect whether predictability holds at the end of the sample, then evidence in Clark and McCracken (2003) suggests that out-of-sample tests are overall more effective. It is important to keep in mind, however, that neither in-sample nor out-of-sample tests of predictability have been designed for the objective of testing end-of-sample predictability.

4.2. Structural Break in α

Throughout this subsection, we will assume that $E(w_t) = 0$ and that the functional central limit holds, i.e.,

$$\begin{bmatrix} T^{-\frac{1}{2}} \sum_{t=1}^{[rT]} w_t u_t \\ T^{-\frac{1}{2}} \sum_{t=1}^{[rT]} w_t \end{bmatrix} \Rightarrow \overline{\Sigma}^{\frac{1}{2}} W(r)$$

where $\overline{\Sigma}$ is a $2k \times 2k$ positive definite matrix and $W(r)$ is the $2k$-dimensional standard Brownian motion. We will analyze the size properties of both in-sample and out-of-sample tests of predictability when the regression model is misspecified under the null hypothesis due to unmodelled structural change. The regression model is

[f]There is one counterexample to this tendency, in which out-of-sample tests will tend to have higher power than in-sample tests: Suppose that the break in β occurs at exactly $[\lambda T]$ where $\lambda = 0.5$. Further suppose that in the first half of the sample $\beta = -c$ and in the second half $\beta = c$ where c is some constant. In that case, the in-sample test will have zero power asymptotically, whereas the out-of-sample test will have some power. This counterexample, however, seems more of an intellectual curiosity because it requires three unrealistic conditions. First, a switch in sign seems unlikely in situations that would suggest the use of a one-sided t-test, as is typically the case in applied work. Second, it is unlikely that the deviations from $\beta = 0$ exactly offset one another. Third, it is unlikely that the break occurs exactly at $[0.5T]$. Even for small deviations from these assumptions the counterexample breaks down.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

misspecified under the null hypothesis if some conditioning variable is incorrectly omitted from v_t when conducting a test of $H_0 : \beta = 0$. A leading example of such a situation is a one-time break in the intercept that is not modelled explicitly. Specifically, we stipulate that the intercept term satisfies in population:

$$\alpha_{1t} = \begin{cases} 0 & t = 1, 2, \ldots, [T/2] \\ 1 & t = [T/2] + 1, \ldots, T \end{cases}.$$

First, we derive the asymptotic null distributions of the in-sample tests under this form of structural change in α. For expository purposes we focus on the t-test. Similar results could be established for the F -test as well. Recall the t-statistic for testing $H_0 : \beta_1 = 0$. Given that

$$t = \sqrt{T}\hat{\beta}_1 / \hat{\sigma}_{\beta_1} \tag{18}$$

where $\hat{\sigma}^2_{\beta_1}$ is the $(l+1) \times (l+1)$ element of $\hat{\sigma}^2((1/T)\sum_{t=1}^{T} x_t x_t')^{-1}$, we will focus on the distribution of the OLS estimator. Substituting for α_1 the value of $1/2$, which is the probability limit of the OLS estimator of α_1 under Assumption 1, we have (with some abuse of notation) that:

$$\begin{aligned} \sqrt{T}(\hat{\gamma} - \gamma) &= \sqrt{T}\left(\sum_{t=1}^{T} x_t x_t'\right)^{-1} \left[\sum_{t=1}^{[T/2]} x_t\left(u_{t+1} - \frac{1}{2}\right) + \sum_{t=[T/2]+1}^{T} x_t\left(u_{t+1} + \frac{1}{2}\right)\right] \\ &= \sqrt{T}\left(\sum_{t=1}^{T} x_t x_t'\right)^{-1} \sum_{t=1}^{T} x_t u_{t+1} - \frac{1}{2}\sqrt{T}\left(\sum_{t=1}^{T} x_t x_t'\right)^{-1} \\ &\quad \times \left[\sum_{t=1}^{[T/2]} x_t - \sum_{t=[T/2]+1}^{T} x_t\right]. \end{aligned} \tag{19}$$

Thus, the limiting null distribution of the t-statistic is normal with mean zero and a variance given by the component of

$$(E(x_i x_i'))^{-1}\left[\sigma^2 E(x_i x_i') + \frac{1}{4}\Delta\right](E(x_i x_i'))^{-1} \Big/ \left(\sigma^2 + \frac{1}{4}\right)$$

that corresponds to β_1. This result obviously differs from the $N(0, 1)$ null limiting distribution in Proposition 1. Therefore, the size of the t-test will be distorted if standard critical values are used.

The result that in-sample tests of predictability based on standard critical values are invalid when the model is misspecified under the null hypothesis is not surprising. What is perhaps less well appreciated is that similar size distortions arise in conducting out-of-sample tests of predictability. For example, consider the GM-test and the

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

ENC-test with $R = [T/2]$. Following the proof of Proposition 3 in the Appendix, one can show that

$$GM = 2\sum_{S=R+1}^{T} \tilde{h}_S' \frac{1}{S}\sum_{s=1}^{S} \tilde{h}_s - \sum_{S=R+1}^{T} \frac{1}{S^2}\sum_{s=1}^{S} \tilde{h}_s' \sum_{u=1}^{S} \tilde{h}_u + o_p(1)$$

$$ENC = \sum_{S=R+1}^{T} \tilde{h}_S' \frac{1}{S}\sum_{s=1}^{S} \tilde{h}_s + o_p(1),$$

where $\tilde{h}_t = E(w_t w_t')^{-\frac{1}{2}} w_t(u_t - 0.5 + I(t > [0.5T]))$. Let $\widetilde{W}_1(r)$ and $\widetilde{W}_2(r)$ be k-dimensional vectors that consist of the first k and last k elements of $\widetilde{\Sigma}W(r)$. Because the limit distribution of the partial sum of $\tilde{h}_t$

$$T^{-\frac{1}{2}}\sum_{t=1}^{[Tr]} \tilde{h}_t \Rightarrow E(w_t w_t')^{-\frac{1}{2}}[\widetilde{W}_1(r) - 0.5\widetilde{W}_2(r)I(r \le 0.5) + (\widetilde{W}_2(r) - \widetilde{W}_2(0.5))I(r > 0.5)],$$

differs from that in the absence of structural change, critical values based on the distributions derived in Propositions 2 and 3 will be invalid and the *GM*- and *ENC*-tests will be size distorted. This argument shows that the *GM*-test and the *ENC*-test based on recursive regressions do not allow for unmodelled structural change in the conditioning variables. Analogous results hold when inference is based on rolling regressions.

Our analysis demonstrates that structural change under the null hypothesis distorts the size of both in-sample and out-of-sample tests of predictability, as currently implemented. This result is driven by the use of global asymptotics. As a referee pointed out, an alternative way of modelling structural change under the null involves the use of local asymptotics. In that case:

$$\alpha_{1t} = \begin{cases} 0 & t = 1, 2, \ldots, [T/2] \\ 1\big/\sqrt{T} & t = [T/2] + 1, \ldots, T \end{cases}$$

The difference to the earlier analysis is that here the structural change vanishes asymptotically. Not surprisingly, under local structural change in α it can be shown that the asymptotic distributions of both in-sample and out-of-sample tests are the same as the null distribution in Propositions 1–3. Details are omitted to conserve space. An immediate implication is that this form of structural change cannot explain the alleged tendency of in-sample tests to reject the null more often than out-of-sample tests when the null is true. Both types of tests are equally reliable under the null. Which of these alternative thought experiments provides a better approximation to the small-sample behavior of tests of predictability is an open question. One would expect that the global asymptotics would provide a better

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

approximation when the magnitude of the structural change is large and the local asymptotics when the magnitude is small. We do not pursue this question. Our main point has been simply to show that there is no presumption that the in-sample test is less reliable in the presence of unmodelled structural change under the null.

5. ANALYSIS IN THE PRESENCE OF DYNAMIC MISSPECIFICATION

Even in the absence of structural change, the null model may be dynamically misspecified. Specifically, the regression errors may be serially correlated under the null. The tests of predictability that we have studied so far are not designed to be robust to dynamic misspecification under the null. This is not a problem to the extent that in many applications the model of interest under the null involves errors that follow a m.d.s. In other applications, however, it is useful to be able to allow for serial correlation in the error term under the null of no predictability. Serially correlated errors under the null arise, for example, in so-called long-horizon regressions of asset returns that involve multi-step ahead projections (see e.g., Mark, 1995; Mark and Choi, 1997; Stock and Watson, 2003). Similar problems arise in predicting inflation rates or rates of economic growth over h periods. In that case, the model is

$$y_{t+h}^h = \gamma' x_t + u_{t+h}^h = \alpha' v_t + \beta' w_t + u_{t+h}^h, \quad h \epsilon \{2, 3, \ldots\} \tag{20}$$

where u_{t+h}^h is serially correlated under the null.

In-sample tests may be made robust to dynamic misspecification by using appropriate standard errors that are robust to serial correlation under the null. The null distribution remains unchanged. In contrast, for the GM test and the encompassing test, the null distribution changes under dynamic misspecification. That null distribution, which has been derived by Clark and McCracken (2004), depends on the data generating process, but may be approximated by bootstrap methods in practice. The implementation of these tests therefore is no more difficult than in the absence of serial correlation.[g] With these modifications both in-sample and out-of-sample tests can be expected to be reliable asymptotically, allowing us to compare their asymptotic power against local alternatives. The next subsection presents the local asymptotic distributions for each test. What makes this case interesting is that, unlike in the standard case, there is no presumption that in-sample tests will be more powerful under these conditions, since the Neyman–Pearson lemma does not apply in the presence of dynamic misspecification.

[g]A related test that is robust against dynamic misspecification has also been proposed by Chao et al. (2001). Yet another test of predictability that allows for dynamic misspecification under the null is presented in Corradi and Swanson (2002), but that paper focuses on the problem of testing equal predictive accuracy of two nested models against the alternative of possibly nonlinear predictability.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

5.1. The Asymptotic Distributions of Misspecification–Robust Tests of Predictability Under Local Alternatives

First, we describe the dynamic-misspecification robust versions of the in-sample tests. For testing H_1, we consider the robust Wald test statistic

$$\textit{robust } W = T(\hat{\beta} - \beta)'\widehat{\Omega}_{\beta}^{-1}(\hat{\beta} - \beta) \tag{21}$$

where $\widehat{\Omega}_{\beta}$ is the lower-right $k \times k$ submatrix of $\widehat{\Omega} = ((1/T)\sum_{t=1}^{T} x_t x_t')^{-1} \times \widehat{\Sigma}((1/T)\sum_{t=1}^{T} x_t x_t')^{-1}$. In the presence of dynamic misspecification, one would estimate Σ by a Newey–West type nonparametric estimator. For testing H_2, we consider the robust t test statistic

$$\textit{robust } t = \sqrt{T}\hat{\beta}_1/\hat{\omega}_{\beta_1} \tag{22}$$

where $\hat{\omega}_{\beta_1}$ is the $(l+1) \times (l+1)$ element of $\widehat{\Omega}$. The following proposition provides the asymptotic distributions of these test statistics under a sequence of local alternatives.

Proposition 7. *Suppose that Assumption* 1 *holds. Under a sequence of local alternatives of the form H_1' and H_2', respectively,*

$$\textit{robust } W \xrightarrow{d} \chi^2_{\delta_3}(k) \tag{23}$$

$$\textit{robust } t \xrightarrow{d} N(\delta_{14}, 1), \tag{24}$$

where $\chi^2_{\delta_1}(k)$ is a noncentral χ^2 random variable with degree of freedom k and noncentral parameters

$$\delta_{13} = [0_{1\times l}c']E(x_t x_t')\Sigma^{-1}E(x_t x_t')[0_{1\times l}c']',$$

$$\delta_{14} = e_{l+1}'\left(1/\omega_{\beta_1}^{1/2}\right)[E(x_t x_t')]^{-1}E(x_t w_{t1})c_1,$$

and ω_{β_1} is the $(l+1) \times (l+1)$th entry of $\Omega = E(x_t x_t')^{-1}\Sigma E(x_t x_t')^{-1}$.

The proof of Proposition 7 is straightforward and thus is omitted. The null limit distribution emerges as a special case of this result with $c = \mathbf{0}$ and $c_1 = 0$, respectively.

Next, we consider dynamic-misspecification robust versions of the recursive GM test and the recursive encompassing test. Except for a scaling factor, the tests are identical to tests proposed by Clark and McCracken (2004) in the context of long-horizon regression tests. We do not normalize the test statistics because the null

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

distributions are not pivotal under any normalization.

$$robust\ GM = \sum_{t=R+1}^{T} (\bar{u}_{0t+h}^{h2} - \bar{u}_{1t+h}^{h2}),$$

$$robust\ ENC = \sum_{t=R+1}^{T} (\bar{u}_{0t+h}^{h2} - \bar{u}_{0t+h}^{h}\bar{u}_{1t+h}^{h}),$$

where the definition of the losses is analogous to Sec. 2. Since these robust out-of-sample tests are no longer pivotal under the null, in practice, the critical values must be computed by bootstrap methods (see Clark and McCracken, 2004). Since here we focus on the asymptotic properties of the tests for a given data generating process, we are able to dispense with the bootstrap.

Proposition 8. *Suppose that Assumptions* 1 *and* 3 *hold. Under a sequence of local alternatives of the form* H_1'

$$\begin{aligned} robust\ GM \xrightarrow{d} & \int_{\pi}^{1} \frac{2}{r} \widetilde{W}'(r) d\widetilde{W}(r) \\ & + 2\delta' \int_{\pi}^{1} \frac{1}{r} \widetilde{W}(r) dr + 2\delta'(\widetilde{W}(1) - \widetilde{W}(\pi)) + 2(1-\pi)\delta'\delta \\ & - \int_{\pi}^{1} \frac{1}{r^2} (\widetilde{W}(r) + r\delta)'(\widetilde{W}(r) + r\delta) dr \end{aligned} \tag{25}$$

$$\begin{aligned} robust\ ENC \xrightarrow{d} & \left[\int_{\pi}^{1} \frac{1}{r} \widetilde{W}'(r) d\widetilde{W}(r) \right. \\ & \left. + \delta' \int_{\pi}^{1} \frac{1}{r} \widetilde{W}(r) dr + \delta'(\widetilde{W}(1) - \widetilde{W}(\pi)) + (1-\pi)\delta'\delta \right], \end{aligned} \tag{26}$$

where $\widetilde{W}(r) = L'E(x_t x_t')^{-1/2}\Sigma^{1/2}W(r)$, *W is a* $(k+l)$*-dimensional standard Brownian motion,* $\delta = L'E(x_t x_t')^{-1/2}E(x_t w_t')c$, *and L is defined in Sec.* 2.

Note that Propositions 7 and 8 allow for serially correlated errors as well as heteroskedastic errors.

5.2. Asymptotic Power Comparison of In-Sample and Out-of-Sample Tests of Predictability

The data generating process is the same as for Sec. 2.2, except that we postulate that $u_{t+h}^h = 0.9u_t^h + \zeta_{t+h}^h$ where $\zeta_{t+h}^h \sim NID(0, \sigma_\zeta^2)$. We set $\sigma_\zeta^2 = 0.0095$, so the variance of the regression error is the same as in Sec. 2.2. We abstract from data mining and from structural change. Figure 3 is constructed analogous to the earlier figures. It shows that the robust W-test is more powerful than the robust out-of-sample tests

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

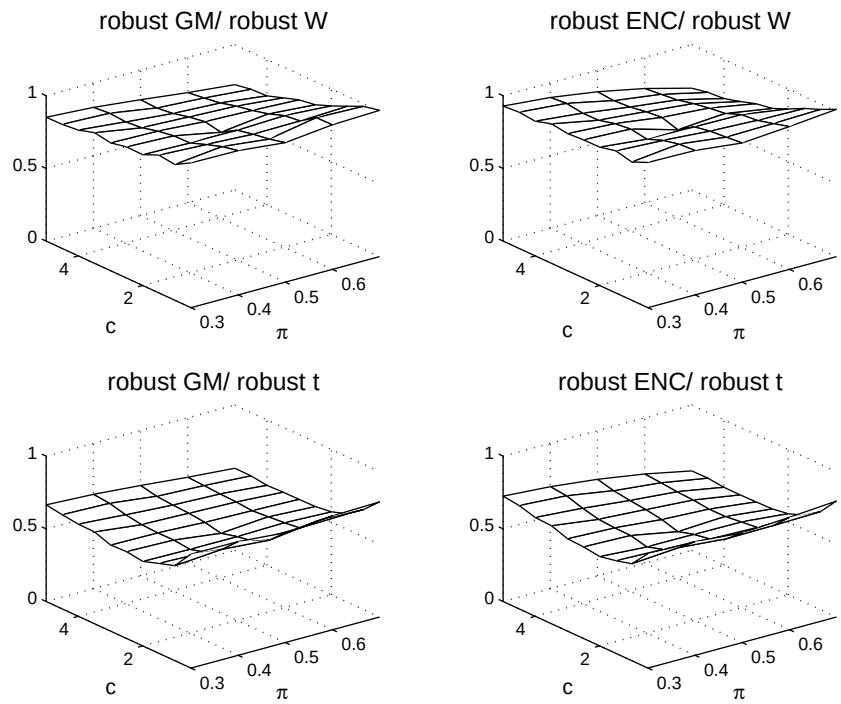

Figure 3. Relative power of predictability tests that are robust against dynamic misspecification.

in most cases. The robust out-of-sample tests in some cases have only 70% of the power of the robust W-test. Only for $c = 0.5$ and $\pi = 0.7$ the out-of-sample tests are marginally more powerful than the robust W-test. The power gain is only about 1% relative to the robust W-test. As expected, the robust t-test is more powerful than the robust out-of-sample tests across the board. The power of the robust out-of-sample tests is between 55% and 80% of the power of the robust t-test. We conclude that even allowing for dynamic misspecification under the null is unlikely to overturn our conclusion that in-sample tests are likely to be more credible than out-of-sample tests in detecting predictability. In fact, the power differences are generally more pronounced than in the standard case.

6. CONCLUSION

It is common for empirical researchers to find significant evidence of in-sample predictability, but no significant out-of-sample predictive relationship. This is true even after controlling for finite-sample distortions. In this paper, we have used asymptotic theory to shed light on this fact. The conventional wisdom is that this empirical tendency reflects the lack of reliability of in-sample tests under the null of no predictability. As a result, there is a tendency to discount evidence in favor

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

of predictability based on in-sample tests. We showed that this interpretation is questionable.

First, we showed that there is no presumption that in-sample tests of predictability suffer from greater size distortions than out-of-sample tests. Our analysis distinguished between environments that are subject to data mining and environments that are free from data mining. We also considered the possibility of unmodelled structural breaks and of dynamic model misspecification under the null hypothesis of no predictability.

Second, we provided an alternative explanation for the tendency of significant in-sample test results to break down out of sample. We showed that in-sample tests in many cases have higher power than out-of-sample tests. Thus, the tendency of in-sample tests to reject more frequently than out-of-sample tests – rather than being a sign of the unreliability of in-sample tests under the null – may be a natural consequence of the null being false. The extent of the power advantages of in-sample tests depends on the environment, in which tests are conducted. It also depends on the precise nature of the hypothesis of interest. Notwithstanding these qualifications, we found that in-sample tests in most cases have higher power than out-of-sample tests of predictability – often by a substantial margin.

Our conclusion is based on an analysis of the local asymptotic power of these tests in a typical, but stylized environment. There are two obvious caveats. First, to the extent that the limit distributions of in-sample and out-of-sample tests of predictability are data-dependent, our results are illustrative examples only. Applied researchers will need to give careful attention to the size and power properties of these tests on a case-by-case basis. Second, many applications involve fairly small samples and asymptotic approximations may not be accurate enough for an assessment of relative power. Given the favorable experience with bootstrap versions of tests of predictability tests in Kilian (1999), Kilian and Taylor (2003), Rapach and Wohar (2003), and Clark and McCracken (2004), we conjecture that in practice bootstrap methods may be used to address both of these concerns.

Finally, we note that our results should not be interpreted as evidence against the use of out-of-sample tests of predictability in general, but against their uncritical use in applied work. Our analysis shows that it is important to be clear about the objective of predictability testing. It will be difficult to find convincing applications of out-of-sample tests of predictability in standard environments such as the settings considered in this paper. There are, however, problems for which in-sample tests are not well suited. For example, there is an important distinction between finite-sample predictability in a real-time setting and predictability that exists in population (see, e.g., Amato and Swanson, 2001; Ashley et al., 1980; Chao et al., 2001). Throughout this paper, we have focused on the ability of tests of predictability to detect the presence of dynamic correlations in population, consistent with a large body of empirical work. Tests designed with this objective in mind, however, may not be useful for detecting whether this predictability may be exploited in real time (even ignoring the problem of data revisions). A further qualification is that out-of-sample tests have been shown to have advantages in detecting predictability that exists only toward the end of the sample (see Clark and McCracken, 2003). It is important to note, however, that neither in-sample nor out-of-sample tests of predictability have been designed with that objective in mind.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

PROOFS

First, we will present some FCLT and stochastic integral results which will be used in the proofs. By Lemma A.4 of Clark and McCracken (2001), Q is idempotent. By Schur's decomposition theorem (Theorem 13 of Magnus and Neudecker, 1999, p. 16), it follows that there is an $l \times k$ matrix L such that $LL' = Q$ and $L'L = I_k$. Let $h_t = L'E(x_t x_t')^{-1/2} x_t(u_t + T^{-1/2} w_t' c)$. Then it follows from the FCLT that

$$T^{-1/2} \sum_{t=1}^{[rT]} h_t \Rightarrow \delta r + L'E(x_t x_t')^{-1/2} \Sigma^{1/2} W(r), \tag{27}$$

where $[x]$ denotes the integer part of x and $\Rightarrow$ denotes the weak convergence in the space of cadlag functions on $[0, 1]$. It further follows from applications of Theorem 2.1 of Hansen (1992), (27) and the continuous mapping theorem that

$$\begin{aligned} \sum_{t=[\pi T]+1}^{T} h_t' \frac{1}{t} \sum_{s=1}^{t} h_s &\xrightarrow{d} \int_{\pi}^{1} \frac{1}{r} \widetilde{W}'(r) d\widetilde{W}(r) \\ &\quad + \delta' \int_{\pi}^{1} \frac{1}{r} \widetilde{W}(r) dr + \delta'(\widetilde{W}(1) - \widetilde{W}(\pi)) + (1 - \pi)\delta'\delta \end{aligned} \tag{28}$$

where $\widetilde{W}(r) = L'E(x_t x_t')^{-1/2} \Sigma^{1/2} W(r)$.

Proofs of Propositions 3 *and* 8. The proof of Proposition 3 is a special case of that of Proposition 8. Following the arguments used in the proofs of (McCracken 1999, Lemma 3.2) and Clark and McCracken (2000, Lemmas A10 and A12), one can show that

$$\begin{aligned} \textit{robust GM} &= 2 \sum_{S=R+1}^{T} \left(\varepsilon_S x_S' + \frac{c' w_S x_S'}{\sqrt{T}} \right) \left(\sum_{s=1}^{S} x_s x_s' \right)^{-1} \left(\sum_{s=1}^{S} x_s \varepsilon_s + \frac{x_s w_s' c}{\sqrt{T}} \right) \\ &\quad - 2 \sum_{S=R+1}^{T} \left(\varepsilon_S v_S' + \frac{c' w_S x_S'}{\sqrt{T}} \right) \left(\sum_{s=1}^{S} v_s v_s' \right)^{-1} \left(\sum_{s=1}^{S} v_s \varepsilon_s + \frac{x_s w_s' c}{\sqrt{T}} \right) \\ &\quad + \sum_{S=R+1}^{T} \sum_{s=1}^{S} \left(\varepsilon_s v_s' + \frac{c' w_s v_s'}{\sqrt{T}} \right) \left(\sum_{s=1}^{S} v_s v_s' \right)^{-1} v_S v_S' \left(\sum_{s=1}^{S} v_s v_s' \right)^{-1} \\ &\quad \times \left(\sum_{s=1}^{S} v_s \varepsilon_s + \frac{v_s w_s' c}{\sqrt{T}} \right) - \sum_{S=R+1}^{T} \sum_{s=1}^{S} \left(\varepsilon_s x_s' + \frac{c' w_s v_s'}{\sqrt{T}} \right) \\ &\quad \times \left(\sum_{s=1}^{S} x_s x_s' \right)^{-1} x_S x_S' \left(\sum_{s=1}^{S} x_s x_s' \right)^{-1} \left(\sum_{s=1}^{S} x_s \varepsilon_s + \frac{x_s w_s' c}{\sqrt{T}} \right) \\ &= 2 \sum_{S=R+1}^{T} \left(\varepsilon_S x_S' + \frac{c' w_S x_S'}{\sqrt{T}} \right) \frac{1}{S} \left\{ \left(E\left(x_t x_t'\right) \right)^{-1} - \begin{bmatrix} (E(v_t v_t'))^{-1} & 0_{l \times k} \\ 0_{k \times l} & 0_{k \times k} \end{bmatrix} \right\} \end{aligned}$$

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

$$
\begin{aligned}
&\times \sum_{s=1}^{S}\left(x_s\varepsilon_s + \frac{x_s w_s' c}{\sqrt{T}}\right) \\
&- \sum_{S=R+1}^{T}\sum_{s=1}^{S}\left(\varepsilon_s x_s' + \frac{c' w_s x_s'}{\sqrt{T}}\right)\frac{1}{S^2}\left\{\left(E\left(x_t x_t'\right)\right)^{-1} - \begin{bmatrix} \left(E\left(v_t v_t'\right)\right)^{-1} & 0_{l\times k} \\ 0_{k\times l} & 0_{k\times k}\end{bmatrix}\right\} \\
&\times \sum_{s=1}^{S}\left(\frac{x_s\varepsilon_s}{\sqrt{T}} + \frac{x_s w_s' c}{T}\right) + o_p(1) \\
&= 2\sum_{S=R+1}^{T} h_S' \frac{1}{S}\sum_{s=1}^{S} h_s - \sum_{S=R+1}^{T}\frac{1}{S^2}\sum_{s=1}^{S} h_s' \sum_{u=1}^{S} h_u + o_p(1), \qquad (29)
\end{aligned}
$$

Thus, (25) in Proposition 8 follows from the continuous mapping theorem, (27) and (29). Similarly,

$$
\textit{robust ENC} = \sum_{S=R+1}^{T} h_S' \frac{1}{S}\sum_{s=1}^{S} h_s + o_p(1). \qquad (30)
$$

The desired result (26) follows from (27), (28), and (30).

Proof of Propositions 4 and 5. Under Assumptions 1 and 3, we have for a sequence of local alternatives

$$
\left[\frac{\sum_{t=1}^{T}\hat{u}_{1,t+1}^2 - \hat{u}_{0,t+1}^2}{\hat{\sigma}_1^2}, \ldots, \frac{\sum_{t=1}^{T}\hat{u}_{M,t+1}^2 - \hat{u}_{0,t+1}^2}{\hat{\sigma}_M^2}\right]' \xrightarrow{d} [\chi_{d_1}^2(k_1), \ldots, \chi_{d_M}(k_M)]' \qquad (31)
$$

Under Assumptions 1 and 4 we have for a sequence of local alternatives

$$
\left[\sqrt{T}\hat{\beta}_{1,T}/\hat{\sigma}_{\beta_1}, \ldots, \sqrt{T}\hat{\beta}_{M,T}/\hat{\sigma}_{\beta_M}\right]' \xrightarrow{d} [v_1, \ldots, v_M]' \qquad (32)
$$

Applications of the continuous mapping theorem to (31) and (32) complete the proof of Propositions 4 and 5, respectively.

Proof of Proposition 6. Let

$$
h_t = L_j'[E(x_{j,t}x_{j,t}')]^{-1/2} x_{j,t}(u_t + T^{-1/2}c(\zeta_j)' w_{j,t})
$$

The proof of Proposition 6 is analogous to the proofs of Propositions 2 and 3 except that (27) is replaced with

$$
T^{-1/2}\sum_{t=1}^{rT}[\, h_{1,t} \quad h_{2,t} \quad \cdots \quad h_{M,t}\,]' \Rightarrow \sigma dr + \Omega^{1/2}B(r) \qquad (33)
$$

and that the continuous mapping theorem is applied to the max functional.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

ACKNOWLEDGMENT

We have benefited from comments at the 2002 European Econometric Society Meeting, the 2002 NBER Summer Institute and the 2002 EC^2 Conference. We also thank seminar participants at Bocconi, Bonn, CORE, the European Central Bank, Exeter, Helsinki, INSEAD, Leuven, Montreal, Pittsburgh, Pompeu Fabra, Southampton, Tokyo Metropolitan, Tokyo, Warwick, Waseda, Yokohama National and York. We especially thank two anonymous referees, Valentina Corradi, Todd Clark, Frank Diebold, Robert Engle, Scott Gilbert, Clive Granger, Alastair Hall, Kirstin Hubrich, Michael McCracken, Peter Reinhard Hansen, Barbara Rossi, Norman Swanson, and Ken West for helpful discussions. Part of this research was conducted while the second author served as an adviser at the European Central Bank (ECB). The views expressed in this paper do not necessarily reflect the opinion of the ECB or its staff.

REFERENCES

Amato, J. D., Swanson, N. R. (2001). The real-time predictive content of money for output. *J. Monetary Econ.* 48:3–24.

Anderson, T. W. (1994). *The Statistical Analysis of Time Series*. New York: Wiley.

Ashley, R., Granger, C. W. J., Schmalensee, R. (1980). Advertising and aggregate consumption: An analysis of causality. *Econometrica* 48:1149–1167.

Berkowitz, J., Giorgianni, L. (2001). Long-horizon exchange rate predictability? *Rev. Econ. Statist.* 83:81–91.

Campbell, J. Y., Shiller, R. J. (1988a). The dividend-price ratio and expectations of future dividends and discount factors. *Rev. Financial Stud.* 1:195–228.

Campbell, J. Y., Shiller, R. J. (1988b). Stock prices, earnings, and expected dividends. *J. Financ.* 43:661–676.

Chao, J. C., Corradi, V., Swanson, N. R. (2001). An out-of-sample test for granger causality. *Macroecon. Dyn.* 5:598–620.

Chinn, M. D., Meese, R. A. (1995). Banking on currency forecasts: How predictable is change in money? *J. Int. Econ.* 38:161–178.

Clarida, R. H., Taylor, M. P. (1997). The term structure of forward exchange premiums and the forecastability of spot exchange rates: Correcting the errors. *Rev. Econ. Statist.* 89:353–361.

Clarida, R. H., Sarno, L., Taylor, M. P., Valente, G. (2003). The out-of sample success of term structure models as exchange rate predictors: A step beyond. *J. Int. Econ.* 60:61–83.

Clark, T. E. (2004). Can out-of-sample forecast comparisons help prevent overfitting? Forthcoming. *J. Forecast* 23:115–139.

Clark, T. E., McCracken, M. W. (2000). Not-for-Publication Appendix to Tests of Equal Forecast Accuracy and Encompassing for Nested Models. Manuscript. Federal Reserve Bank of Kansas City and Louisiana State University.

Clark, T. E., McCracken, M. W. (2001). Tests of equal forecast accuracy and encompassing for nested models. *J. Econometrics.* 105:85–110.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

Clark, T. E., McCracken, M. W. (2003). The power of tests of predictive ability in the presence of structural breaks. Forthcoming. *J. Econometrics.*

Clark, T. E., McCracken, M. W. (2004). Evaluating long-horizon forecasts. Manuscript. Department of Economics, University of Missouri at Columbia.

Corradi, V., Swanson, N. R. (2002). A consistent test for nonlinear out of sample predictive accuracy. *J. Econometrics.* 110:353–381.

Corradi, V., Swanson, N. R., Olivetti, C. (2001). Predictive ability with cointegrated variables. *J. Econometrics* 104:315–358.

Dasgupta, N., Spurrier, J. D. (1997). A class of multivariate chi-square distributions with applications to comparisons with a control. *Commun. Statist. Theory Method* 26:1559–1573.

Denton, F. T. (1985). Data mining as an industry. *Rev. Econ. Statist.* 67:124–127.

Diebold, F. X., Mariano, R. S. (1995). Comparing predictive accuracy. *J. Business Econ. Statist.* 13:253–263.

Fama, E. F., French, K. R. (1988). Dividend yields and expected stock returns. *J. Financial Econ.* 22:3–25.

Faust, J., Rogers, J., Wright, J. (2003). Exchange rate forecasting: The errors we've really made. *J. Int. Econ.* 60:35–59.

Gilbert, S. (2001). Sampling Schemes and Tests of Regression Models. Manuscript. Department of Economics, Southern Illinois University at Carbondale.

Goetzmann, W. N., Jorion, P. (1993). Testing the predictive power of dividend yields. *J. Finance* 48:663–679.

Goetzmann, W. N., Jorion, P. (1995). A longer look at dividend yields. *J. Business* 68:483–508.

Granger, C. W. J. (1990). *Modeling Economic Time Series: Readings in Econometric Methodology*. Oxford, UK: Oxford University Press.

Hansen, B. E. (1992). Convergence to stochastic integrals for dependent and heterogeneous processes. *Econometric Theory* 8:489–500.

Hansen, P. R. (2000). The Distribution of the Maximal R^2. Manuscript. Department of Economics, University of California at San Diego.

Hansen, P. R. (2001). The Reality Check for Data Snooping: A Comment on White. Manuscript. Department of Economics, Brown University.

Harvey, D. I., Leybourne, S. J., Newbold, P. (1998). Tests for forecast encompassing. *J. Business Econ. Statist.* 16:254–259.

Inoue, A., Kilian, L. (2003). On the Selection of Forecasting Models. Working Paper No. 214. European Central Bank.

Kilian, L. (1999). Exchange rates and monetary fundamentals: What do we learn from long-horizon regressions? *J. Appl. Econometrics* 14:491–510.

Kilian, L., Taylor, M. P. (2003). Why is it so difficult to beat the random walk forecast of exchange rates? *J. Int. Econ.* 6:85–107.

Magnus, J. R., Neudecker, H. (1999). *Matrix Differential Calculus with Applications in Statistics and Econometrics*. Revised edition, Chichester, UK: Wiley.

Mark, N. C. (1995). Exchange rates and fundamentals: Evidence on long-horizon predictability. *Amer. Econ. Rev.* 85:201–218.

Mark, N. C., Choi, D. (1997). Real exchange rate prediction over long horizons. *J. Int. Econ.* 43:29–60.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

ORDER REPRINTS

McCracken, M. W. (1999). Asymptotics for Out of Sample Tests of Causality. Manuscript. Department of Economics, Louisiana State University.

McCracken, M. W. (2001). Data Mining and Out-of-Sample Inference. Manuscript. Department of Economics, Louisiana State University.

Meese, R. A., Rogoff, K. (1983). Empirical exchange rate models of the seventies: Do they fit out-of-sample? *J. Int. Econ.* 14:3–24.

Meese, R. A., Rogoff, K. (1988). Was it real? The exchange rate-interest differential relation over the modern floating-rate period. *J. Finance* 43:933–948.

Meese, R. A., Rose, A. K. (1991). An empirical assessment of non-linearities in models of exchange rate determination. *Rev. Econ. Stud.* 58:603–619.

Rapach, D. E., Wohar, M. E. (2002). In-Sample vs. Out-of-Sample Tests of Stock Return Predictability in the Context of Data Mining. Manuscript. University of Nebraska.

Rapach, D. E., Wohar, M. E. (2003). Valuation ratios and long-horizon stock price predictability. Forthcoming. *J. Appl. Econometrics.*

Rossi, B. (2003). Testing long-horizon predictive ability with high persistence and the Meese-Rogoff puzzle. Forthcoming: International Economic Review.

Royen, T. (1984). Multivariate comparisons of polynomial distributions. *Biometrical J.* 26:319–332.

Royen, T. (1997). Multivariate gamma distributions (Update). In: Kotz, S., ed. *Encyclopedia of Statistical Sciences.* Vol. 1. New York: Wiley, pp. 419–425.

Stock, J. H., Watson, M. W. (1999). Forecasting inflation. *J. Monetary Econ.* 44:293–335.

Stock, J. H., Watson, M. W. (2003). Forecasting output and inflation: The role of asset prices. *J. Econ. Literature* 41:788–829.

Sullivan, R., Timmermann, A., White, H. (2001). Dangers of data mining: The case of calendar effects in stock returns. *J. Econometrics* 105:249–286.

West, K. D. (1996). Asymptotic inference about predictive ability. *Econometrica* 64:1067–1084.

West, K. D. (2001). Forecasts depend on estimated regression parameters. *J. Business Econ. Statist.* 19:29–33.

White, H. (2000). A reality check for data snooping. *Econometrica* 68:1097–1126.

Copyright © Marcel Dekker, Inc. All rights reserved.

Marcel Dekker, Inc.
270 Madison Avenue, New York, New York 10016

Request Permission or Order Reprints Instantly!

Interested in copying and sharing this article? In most cases, U.S. Copyright Law requires that you get permission from the article's rightsholder before using copyrighted content.

All information and materials found in this article, including but not limited to text, trademarks, patents, logos, graphics and images (the "Materials"), are the copyrighted works and other forms of intellectual property of Marcel Dekker, Inc., or its licensors. All rights not expressly granted are reserved.

Get permission to lawfully reproduce and distribute the Materials or order reprints quickly and painlessly. Simply click on the "Request Permission/ Order Reprints" link below and follow the instructions. Visit the U.S. Copyright Office for information on Fair Use limitations of U.S. copyright law. Please refer to The Association of American Publishers' (AAP) website for guidelines on Fair Use in the Classroom.

The Materials are for your personal use only and cannot be reformatted, reposted, resold or distributed by electronic means or otherwise without permission from Marcel Dekker, Inc. Marcel Dekker, Inc. grants you the limited right to display the Materials only on your personal computer or personal wireless device, and to copy and download single copies of such Materials provided that any copyright, trademark or other notice appearing on such Materials is also retained by, displayed, copied or downloaded as part of the Materials and is not removed or obscured, and provided you do not edit, modify, alter or enhance the Materials. Please refer to our Website User Agreement for more details.

Request Permission/Order Reprints

Reprints of this article can also be ordered at

http://www.dekker.com/servlet/product/DOI/101081ETC200040785